Internship Report

Strategic Planning Department

Statistics Section

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Internship Period: May–July 2026

Declaration

I, **Almuzahim Naseef Muhamad**, declare that this report is my original work, prepared from records that I have been keeping while at Uganda Civil Aviation Authority during my placement between May 4th and July 31st 2026. This document is therefore presented to Uganda Christian University. I affirm that all information contained herein is accurate to the best of my knowledge and that due acknowledgement has been made where external input was received.

Student Signature

_____

Almuzahim Naseef Muhamad

Date: _____

Acknowledgement

I am deeply grateful to the Uganda Civil Aviation Authority (UCAA) for granting me the opportunity to undertake my internship. This placement provided invaluable exposure to aviation statistics and strategic planning processes and allowed me to apply and refine my academic knowledge in a professional environment.

I sincerely thank my supervisors, **Mr. Olipan Jonathan**, Senior Strategic Planning Officer (Statistics), and **Mr. Mukwena David**, my immediate field supervisor, for their guidance, encouragement, and constructive feedback throughout the internship.

Special thanks go to my colleagues and fellow interns, whose collaboration and peer support made the internship both productive and enjoyable. Their willingness to share knowledge and assist in daily tasks fostered a spirit of teamwork that was essential to completing assignments successfully.

Finally, I acknowledge my family and friends for their unwavering support and encouragement during this period. Their motivation and understanding enabled me to remain focused and committed to achieving the objectives of the internship.

Approval

This internship report has been submitted for examination with the approval of:

Student Signature

_____

Almuzahim Naseef Muhamad

Date: _____

Field Supervisor

_____

Date: _____

Manager Human Resource(Training and Development)

_____

Date: _____

Executive Summary

This report presents a comprehensive account of my internship conducted from 4 May 2026 to 31 July 2026. The internship provided practical exposure to aviation statistics, encompassing the collection, entry, reconciliation, and reporting of passenger and cargo data. It also offered opportunities to contribute to strategic initiatives such as updating the Authority's Business Plan.

The experience unfolded in stages. Initial weeks focused on orientation, familiarization with aviation codes, and basic data entry. Subsequent weeks emphasized reconciliation, consolidation of weekly reports into monthly datasets, and preparation of sector passenger and cargo forms, including upcountry statistics for Kasese and Gulu. Later assignments involved integration of statistical insights into the Business Plan for FY 2024/25. The internship concluded with collaborative work on a facilitation proposal addressing operational challenges.

Through this placement, I strengthened technical skills in Excel and Python, developed professional competencies in teamwork, communication, and mentoring, and gained exposure to monitoring and evaluation practices. Despite challenges such as repetitive manual workloads, semi-structured ADHOCS manifests, and coordination among interns, solutions were found through automation, validation techniques, and collaborative shift work.

Overall, the internship was transformative, bridging academic knowledge with real-world aviation practice. The recommendations provided in this report aim to enhance UCAA's data workflows through digitization, automation, and improved facilitation for interns and staff, ensuring that aviation statistics continue to support effective planning and decision-making.

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List of Abbreviations

UCAA	Uganda Civil Aviation Authority
IATA	International Air Transport Association
ICAO	International Civil Aviation Organization
ADHOCS	Unscheduled Flights
M&E	Monitoring and Evaluation
Pax	Passengers
DAS	Daily Aviation Service
FY	Financial Year
AODB	Airport Operational Database
MONUSCO	United Nations Organization Stabilization Mission in the Democratic Republic of the Congo
DCA	Directorate of Civil Aviation

Chapter 1 Introduction

Internships are a core component of university programmes, designed to bridge classroom theory with practical workplace experience. This report documents my placement at the Uganda Civil Aviation Authority (UCAA). The placement provided sustained exposure to the operational processes that support aviation statistics and offered opportunities to apply data science techniques to real institutional problems.

The placement had three main objectives. First, to gain practical workplace experience in data science by engaging with real datasets, workflows, and reporting requirements. Second, to apply and refine technical skills in data collection, cleaning, and analysis using tools such as Excel and Python. Third, to cultivate professional competencies in teamwork, communication, documentation, and problem-solving, enabling effective collaboration within a busy organizational unit.

These general objectives were realized through aviation statistics tasks. My role involved handling air traffic data end-to-end: receiving manifests and ADHOCS from handlers and regional stations, standardizing and entering records into departmental templates, reconciling inconsistencies, and producing and analyzing weekly and monthly reports for management. Over time, the placement expanded beyond routine entry and reconciliation to include the preparation of sector passenger and cargo forms, as well as contributions to strategic assignments such as updating the Authority's Business Plan for FY 2024/25. These higher-level tasks illustrated how operational statistics feed into planning and resource allocation at the Authority.

This report is structured to reflect both the operational and developmental aspects of the internship. Chapter 2 situates the placement within UCAA's institutional context and describes the Statistics Section's role in the Authority's planning cycle. Chapter 3 synthesizes the day-to-day activities into a thematic narrative that highlights progression from induction to strategic contributions. Chapters on skills and knowledge gained describe the technical methods and professional behaviours developed during the placement, while a dedicated chapter examines challenges encountered and the mitigations applied. The report closes with a summary of achievements, prioritized recommendations for improving data workflows, and personal reflections. Selected weekly reports, code listings, sample forms, and supporting figures are provided in the appendices for completeness and verification.

Chapter 2 Organizational Background

2.1 Background of the Authority

Civil aviation in Uganda has a long and evolving history. Air transport began with flying boats landing at Port Bell, Luzira on Lake Victoria. The Directorate of Civil Aviation (DCA) was formed in 1946, and Entebbe Airport was constructed in 1947 as the country's first major aviation facility. Entebbe International Airport was officially commissioned in 1952 by Princess Elizabeth of England, marking Uganda's entry into international civil aviation. The airport's infrastructure continued to evolve, with the current terminal building completed in 1972 to accommodate growing passenger and cargo traffic.

Following the establishment of the East African Community, civil aviation services were managed regionally under the DCA and other government departments. However, fragmentation and inefficiencies in this arrangement eventually led to the creation of the Uganda Civil Aviation Authority (UCAA) in 1991. Since then, UCAA has overseen major modernization projects, including runway extensions, installation of advanced navigation aids, and expansion of passenger facilities. These developments have positioned Entebbe International Airport as a regional hub, while also supporting the growth of upcountry aerodromes such as Gulu, Arua, and Mbarara.

Today, the Authority operates under the Ministry of Works and Transport, providing oversight, regulation, and development of civil aviation in Uganda. Its role extends beyond infrastructure management to include safety oversight, air navigation services, and facilitation of international standards in aviation. Through continuous investment in airport construction and modernization, UCAA has ensured that Uganda's aviation sector remains aligned with global best practices and capable of supporting tourism, trade, and national development.

2.2 Vision of the Authority

The vision of UCAA is to provide the safest, most efficient, and affordable air transport services. This vision reflects the Authority's aspiration to position Uganda as a hub of reliable and accessible aviation in the region.

2.3 Mission of the Authority

The mission of UCAA is to maintain the highest standards of safety, security, and service in civil aviation. This mission underscores the Authority's commitment to international best practices and its role in safeguarding passengers, cargo, and aviation infrastructure.

2.4 Objectives of the Authority

UCAA's objectives emphasize promoting the safe, secure, regular, and efficient use and development of civil aviation both within Uganda and internationally. These objectives guide the Authority's operations, ensuring that aviation services contribute to national development, trade, and tourism.

2.5 Core Values of UCAA

The Uganda Civil Aviation Authority is guided by the following core values:

- **Customer Focus** – Commitment to quality service and passenger satisfaction.
- **Accountability** – Transparency and responsibility in all operations.
- **Teamwork** – Collaboration across departments to achieve common goals.
- **Efficiency** – Streamlined processes for effective service delivery.
- **Innovation** – Adoption of modern technologies and continuous improvement.
- **Integrity** – Upholding ethical standards and compliance with regulations.

2.6 Leadership Structure

UCAA is led by a Director General, supported by Deputy Directors and departmental heads who provide policy and strategic direction. The Strategic Planning Department falls under the corporate division, which is central to this leadership. Within this department, the Statistics Section is responsible for collecting, validating, and reporting aviation statistics. The section's outputs include weekly and monthly air traffic reports, sector passenger and cargo summaries, and upcountry airport reports. Interns and staff report to section supervisors, ensuring accountability and clear lines of communication.

The Strategic Planning Department is headed by the Manager, **Mr. Bukenya Joseph**, supported by a Principal Officer. The Principal oversees managers of the Planning, Statistics, and Monitoring & Evaluation (M&E) sections.

2.7 Organizational Structure

The organizational structure of UCAA situates the Strategic Planning Department alongside operational and regulatory divisions, enabling a two-way flow of information between frontline units and senior management. The Statistics Section is nested within this structure, linking operational data collection to strategic planning. Figure 2.1 illustrates the placement of the Statistics Section within the Authority's hierarchy.

By situating my internship within this organizational framework, I came to appreciate how routine statistical tasks contribute to broader institutional objectives. The Statistics Section is not merely a record-keeping unit; it is a vital component of UCAA's planning machinery, ensuring that decisions are grounded in accurate and timely data.

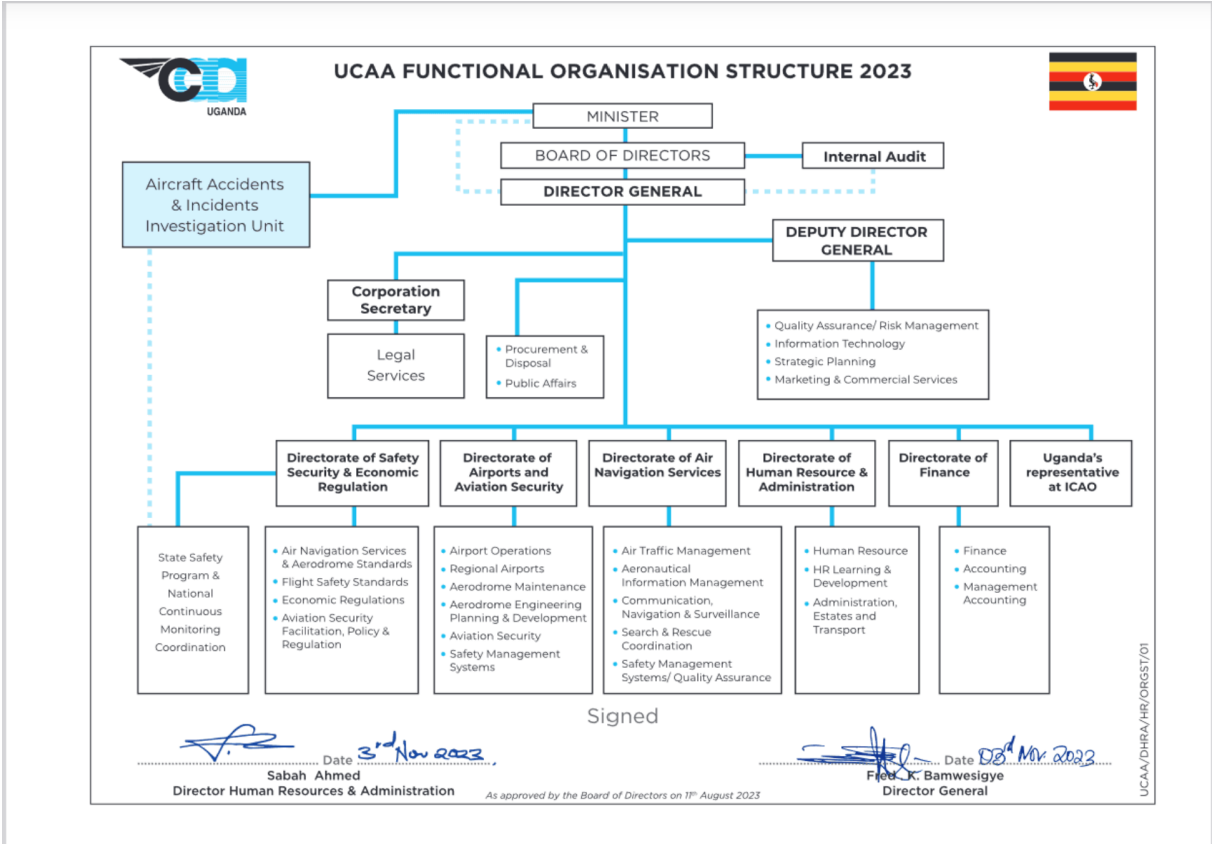


Figure 2.1: Uganda Civil Aviation Authority organizational structure.

Source: Civil Aviation Authority Uganda (https://caa.go.ug)

Chapter 3 Internship Activities

This chapter presents a comprehensive account of the activities undertaken during my internship. It is deliberately detailed, as it reflects the practical work carried out over three months. While many activities were repeated weekly, such as data collection, entry, and reconciliation, these repetitions were essential to mastering the workflow between Strategic Planning, the Control Tower, and handlers. The chapter also highlights advanced reporting formats, strategic contributions, and technical reflections on the systems used.

3.1 Orientation

The internship began with a week of orientation designed to introduce interns to the Uganda Civil Aviation Authority (UCAA) environment. During this period, I was formally welcomed to the Strategic Planning Department and briefed on the Authority's mandate, organizational structure, and departmental workflows.
The orientation included:

- Introduction to staff members and supervisors within the Statistics Section.
- Familiarization with aviation phonetics, ICAO standards, and reporting formats used in weekly and monthly traffic reports.
- Guided walkthrough of the data collection and reconciliation process, including how passenger and cargo statistics are compiled from various sources.

This stage provided the foundation for subsequent tasks in data entry, analysis, and reporting. It also emphasized the importance of accuracy, timeliness, and compliance with international aviation standards in statistical reporting. The phonetic alphabet and ICAO region map are included in Appendix C.

3.2 Data Collection

Data collection was conducted at the beginning of each week through engagement with centers such as DAS, MENZIES, and MONUSCO, where flight data was obtained from logbooks. Handled flights were recorded in these logbooks, while non-handled flights were documented on

ADHOCS. ADHOCS refer to unscheduled flights, with data captured on manifests showing cargo, passengers, and crew details. This process required careful organization to ensure completeness and accuracy.

3.3 Data Entry

Once collected, manifests were entered into departmental templates. Handled flights were relatively straightforward, as their log sheets followed consistent formats. ADHOCS, however, demanded extra effort to standardize before entry. Accuracy was paramount: even small errors could distort passenger and cargo statistics. Over time, I developed speed and precision in entering data, aided by memorization of IATA and ICAO codes.The ICAO code of Entebbe International Airport is **HUEN**. The code follows an order where the The first letter is where the country is located, second letter is the country itself and last two where the airport is located. These codes standardized airport identifiers, ensuring interoperability across global datasets. Refer to Figure C.1 in Appendix C for a sample of the ICAO global map.

3.4 Data Verification

Reconciliation was the most demanding and repetitive task. The Airport Operational Database (AODB) was used for handled flights, but it had significant limitations. It could not process non-handled flights, and its manual nature required approving or declining flights one by one. If the difference between passenger counts from handlers and the Control Tower exceeded two, the flight was declined and feedback sent back to handlers (e.g., “check inbound passenger numbers”). This process was time-consuming and prone to delays. I reflected that with modifications, such as embedding IF statements into the workflow, the process could be semi-automated, reducing manual approvals and improving efficiency.

3.5 Data Analysis

Beyond data collection and verification, I carried out exploratory data analysis on both weekly and monthly Excel datasets. Using Python in VS Code, I aggregated passenger and cargo counts, identified the busiest operators and routes, and calculated punctuality performance based on arrival delays. These tasks reflected concepts from *Introduction to Data Science* and *Data Storytelling and Visualization*, where raw datasets are transformed into meaningful insights.

In Excel, I applied pivot tables to summarize traffic flows and created charts that were later incorporated into the Business Plan. This complemented the Python scripts by providing clear visualizations for non-technical stakeholders.
Representative code snippets are included in Appendix B, Listings B.3 and B.2, which demonstrate how automation and visualization techniques were applied to aviation datasets. The Excel charts used in the Business Plan are referenced in Appendix D. Figures C.5 and C.4 in Appendix C show analysis on weekly reports.

3.6 Consolidation of Reports

Weekly reports were consolidated into monthly datasets. For example, the May air traffic report combined the periods of 8–13, 14–20, 21–27, and 28–31 May. Consolidation required careful validation to avoid duplication or omission. I practiced using Python's `glob()` function to concatenate Excel files, improving reproducibility and reducing manual workload. This experience demonstrated the importance of automation in statistical reporting. Implementation of the `glob()` function is included in Appendix B Listing B.1.

3.7 Sector Passenger (Pax) and Cargo Forms

From Week 7 onward, I worked on sector passenger (Pax) forms, which aggregate inbound and outbound flows by route. These forms revealed traffic distribution, with routes such as Entebbe–Nairobi consistently busy. Cargo forms were derived from the sector Pax datasets, focusing on imports and exports. Preparing upcountry reports for Kasese and Gulu airports added complexity, as regional datasets often required additional reconciliation. By Week 9, I applied Python aggregation to align cargo and route data, improving efficiency since pivot tables were somewhat slow. Figure C.6 is the sector Pax template and is included in Appendix C.

3.8 Form C – Passenger Movement by Aircraft Type

Form C tracks passenger movement by aircraft type. It provides detailed statistics on capacity available, payload, and revenue traffic by flight stage. This form allowed me to analyze Uganda Airlines operations more precisely, correcting discrepancies in passenger counts.This form is handed in monthly to ICAO by the organization .Figure C.7 in Appendix C illustrates a sample Form C entry.

3.9 Form B – Quarterly Passenger Flows

Form B records passenger flows on a quarterly basis, regardless of aircraft type. It complements Form C by providing a broader view of traffic trends over time. This form was useful for identifying seasonal variations and long-term growth patterns.Unlike Form C,it is submitted at the end of the year. Figure C.8 in Appendix C shows a sample Form B entry.

3.10 Business Plan Update

A major assignment was updating the Business Plan for FY 2024/25. The organization has three major plans that include **Master Plan** that covers a period of twenty years, **Business Plan** that covers a period of one year and **Strategic Plan** that covers a five year period. We were told to update the business plan which showed forecasts and goals set by the organization. For this, a long-term air traffic datasets (1992–2025) was used where air traffic statistics across multiple financial years were analyzed, and forecasts for FY 2025/26 and FY 2026/27 were produced. Statistical insights were integrated into the plan, demonstrating how operational statistics feed into strategic planning and national development. Excerpts from the updated Business Plan are included in Appendix D.

3.11 Facilitation Proposal

I was requested by an officer at the facilitation desk to draft a proposal aimed at improving workflow efficiency, given the heavy reliance on manual processes. This proposal remains under development and is therefore not highlighted in detail within this document.

3.12 Report Compilation (Weeks 11–12)

The final two weeks were dedicated to drafting, editing, and finalizing this report. I integrated weekly logs, appendices, and supporting visuals, and consulted with my supervisor for feedback and sign-off. This phase emphasized documentation, reflection, and professional presentation.

Chapter 4 Skills and Knowledge Gained

The internship provided a unique opportunity to apply theoretical knowledge in a professional environment. Over the course of twelve weeks, I developed a wide range of technical, analytical, and professional skills. This chapter outlines the competencies acquired, grouped into technical skills, analytical skills, theory-to-practice connections, and professional development.

4.1 Technical Skills

4.1.1 Data Collection and Entry

Through repeated engagement with handlers (Menzies, DAS, MONUSCO) and the Control Tower, I mastered the process of collecting manifests and ADHOCS. I learned to standardize semi-structured ADHOCS before entry, ensuring consistency. Memorization of IATA and ICAO codes improved efficiency in manual entry and reduced errors.

4.1.2 Data Verification

Verification tasks strengthened my ability to detect and correct inconsistencies. I gained hands-on experience with the Airport Operational Database (AODB), understanding its strengths and limitations. The manual approval process taught me patience and precision, while reflecting on possible automation sharpened my problem-solving mindset.

4.1.3 Programming

I applied Python tools such as the `glob()` function to merge Excel files, reducing manual workload. Later, I used Python aggregation to reconcile cargo and route data, demonstrating how programming can complement traditional spreadsheet methods. These experiences enhanced my confidence in using Python for data science tasks. Code snippets showing the use of `glob()` for merging and functions used to summarize operators' inbound and outbound routes are included in Appendix B.

4.2 Analytical Skills

4.2.1 Sector Passenger and Cargo Analysis

Working on sector pax and cargo forms improved my ability to aggregate and interpret traffic flows. I learned to identify busy routes, seasonal variations, and trade patterns. Upcountry reports for Kasese and Gulu airports expanded my perspective on regional aviation dynamics.

4.2.2 Advanced Reporting Formats

Form C and Form B introduced me to aircraft-specific and quarterly reporting. These formats required precision in verification and broadened my analytical scope from monthly datasets to long-term trends. Correcting Uganda Airlines data reinforced the importance of accuracy in institutional reporting.

4.2.3 Strategic Planning Contributions

Updating the business plan for FY 2024/25 allowed me to integrate statistical insights into organizational strategy. I prepared long-term datasets, analyzed budget allocations, and contributed to evidence-based planning.

4.3 Theory-to-Practice Connections

The internship provided a practical platform to apply concepts from my undergraduate coursework. From *Introduction to Data Science*, I applied data cleaning techniques when standardizing ADHOCS and merging weekly reports. The module on *Data Storytelling and Visualization* was reflected in the preparation of weekly and monthly reports for analysis, as well as in the use of figures to communicate traffic flows clearly to non-technical stakeholders. Finally, *Data Structures* such as dictionaries were directly applied to summarize operator routes and inbound/outbound flows, demonstrating how abstract programming concepts can be transformed into operational aviation statistics.

These connections illustrate how classroom theory translated into professional practice.

4.4 Professional Development

4.4.1 Teamwork and Collaboration

Working within the Statistics Section required constant collaboration. Data collection from handlers and reconciliation with the Control Tower were team-based processes, demanding

coordination and mutual support. The introduction of shift work in later weeks further emphasized the importance of collaboration, as interns had to divide tasks efficiently to avoid duplication. Peer teaching, where I guided new interns through data entry and reconciliation, reinforced my ability to work in teams and highlighted the value of knowledge sharing.

4.4.2 Communication Skills

Professional communication was central to the internship. Providing feedback to handlers during reconciliation — for example, requesting them to "check inbound passenger numbers" when discrepancies arose — required clarity and professionalism. Drafting weekly and monthly reports demanded concise writing. Contributing to the business plan and facilitation proposal further strengthened my ability to communicate complex statistical insights in an acceptable manner.

4.4.3 Adaptability and Problem-Solving

The internship environment was dynamic, with frequent challenges such as semi-structured ADHOCS, manual AODB approvals, and inconsistent datasets. Adapting to these challenges required flexibility and problem-solving skills. For instance, when Excel proved insufficient for a certain task, I explored Python as an alternative. This adaptability ensured that tasks were completed despite obstacles and reinforced resilience in professional practice.

4.4.4 Time Management

Balancing multiple responsibilities — from routine data entry to strategic assignments like the business plan update demanded effective time management. Public holidays and lighter weeks required careful planning to ensure continuity. Coordinating with fellow interns during shift work also required discipline to meet deadlines without compromising quality. These experiences improved my ability to prioritize tasks and manage workload efficiently.

4.4.5 Professional Ethics and Responsibility

The internship emphasized accountability and integrity. Handling aviation statistics required confidentiality and accuracy, as errors could affect institutional decisions. I learned to take responsibility for my outputs, ensuring that reports were reliable and reproducible. Exposure to UCAA's core values customer focus, accountability, teamwork, efficiency, innovation, and integrity reinforced the importance of ethical conduct in professional environments.

Chapter 5 Challenges

No internship is without obstacles, and my placement presented several challenges. These challenges were both technical and organizational, reflecting the complexity of aviation data management. This chapter outlines the major difficulties encountered, focusing on data collection, data entry, system limitations, and coordination.

5.1 Data Collection Challenges

Collecting manifests from handlers such as Menzies, DAS, and MONUSCO was not always straightforward. Handled flights were well documented, but non-handled flights relied on ADHOCS, which were semi-structured and often incomplete. Upcountry reports from airports such as Kasese and Gulu added further complexity, as we were required to re-calculate values to make sure they were right.

5.2 Data Entry Challenges

Data entry demanded precision, yet the repetitive nature of the task made it prone to human error. Memorizing IATA and ICAO codes improved efficiency, but mistakes in code entry could distort entire reports such as the Sector Pax. Balancing speed with accuracy was a constant challenge, especially when deadlines for weekly and monthly reports were tight.

5.3 Reconciliation Challenges

Reconciliation was the most demanding aspect of the internship. The Airport Operational Database (AODB) was used for handled flights, but it had significant limitations. It could not process non-handled flights, and its manual nature required approving or declining flights one by one. If the difference between passenger counts from handlers and the Control Tower exceeded two, the flight had to be declined and feedback sent back to handlers. This process was time-consuming and repetitive, often delaying report finalization. I reflected that with modifications, such as embedding IF statements into the workflow, the process could be semi-automated, reducing manual approvals and improving efficiency.Example of idea for the system can be seen in Appendix B

5.4 System Limitations

Excel was the primary tool for data entry, but it struggled with large datasets . Importing files often required additional handling to align formats. While Python provided more flexibility, integrating it into the workflow required extra effort and was not yet institutionalized. This highlighted the gap between manual processes and modern data science tools.

5.5 Coordination and Workload

The growing number of interns required shift work, which introduced coordination challenges. Ensuring that tasks were not duplicated and that outputs remained consistent demanded clear communication and teamwork. Balancing peer teaching with personal responsibilities also required effective time management.

5.6 Strategic Assignments

Updating the business plan was conceptually demanding. They required integrating statistical insights into organizational strategy, linking aviation data to broader economic planning, and balancing technical detail with strategic objectives.

Chapter 6 Achievements

The internship was not only a learning experience but also an opportunity to make tangible contributions. This chapter outlines the major achievements accomplished during the three-month placement, grouped into operational outputs, analytical contributions, strategic inputs, and professional growth.

6.1 Operational Achievements

6.1.1 Data Collection,entry and analysis

I successfully collected and entered passenger and cargo manifests from handlers and the Control Tower on a weekly basis and also analyzed the data . My memorization of IATA and ICAO codes improved efficiency and reduced errors in manual entry.

6.1.2 Validation

I played an active role in reconciling discrepancies between handler data and Control Tower records. Using the Airport Operational Database (AODB), I applied approval rules and provided feedback to handlers when discrepancies exceeded thresholds. This ensured that weekly and monthly reports were accurate.

6.2 Analytical Achievements

6.2.1 Sector Passenger (Pax) and Cargo Forms

I prepared sector pax forms that aggregated inbound and outbound flows by route, providing insights into traffic distribution. Cargo forms derived from these datasets highlighted trade flows and busy cargo routes. Upcountry reports for Kasese and Gulu airports expanded coverage to regional aviation activity.

6.2.2 Advanced Reporting Formats

I completed Form C, which tracks passenger movement by aircraft type, and initiated Form B, which records quarterly passenger flows. These formats broadened reporting scope from monthly datasets to aircraft-specific and quarterly analysis, strengthening institutional monitoring.

6.3 Strategic Achievements

6.3.1 Business Plan Update

I contributed to updating the business plan for FY 2024/25 and analyzing budget allocations to the transport sector. My inputs ensured that statistical evidence was integrated into organizational strategy.

6.4 Professional Achievements

6.4.1 Teamwork and Peer Teaching

I guided new interns through data entry and reconciliation processes, reinforcing my own understanding while supporting their integration. Shift work required coordination and communication, which I managed effectively to ensure consistent outputs.

6.4.2 Documentation and Reporting

I compiled weekly and monthly reports, ensuring examiner-friendly presentation. The final internship report integrated logs, appendices, and visuals, reflecting professional documentation standards.

Chapter 7 Recommendations

Based on the experiences and challenges encountered during my internship, several recommendations can be made to improve efficiency, accuracy, and strategic impact. These recommendations address technical processes, workflow management, and organizational practices.

7.1 Technical Recommendations

7.1.1 Automation of Reconciliation

The Airport Operational Database (AODB) should be enhanced with automation features. Currently, reconciliation requires manual approval or rejection of flights when discrepancies exceed two passengers. Embedding conditional logic (e.g., IF statements) or integrating Python scripts could automate this process, reducing delays and minimizing human error.

7.1.2 Standardization of ADHOCS

Non-handled flight manifests (ADHOCS) should be standardized into a consistent template. This would reduce the time spent organizing semi-structured data before entry and improve reliability. Training handlers to adopt uniform formats would further streamline reporting.

7.1.3 Integration of Python Workflows

While Excel remains the primary tool, Python offers greater flexibility for large datasets and reconciliation tasks. Institutionalizing Python workflows such as aggregation, merging, and validation scripts would improve reproducibility and reduce manual workload.

7.2 Workflow Recommendations

7.2.1 Enhanced Coordination Between Units

Closer coordination between the Statistics Section, Control Tower, and handlers would improve data accuracy. Establishing clear communication protocols for feedback (e.g., standardized error reports) would reduce delays in reconciliation and ensure timely corrections.

7.2.2 Capacity Building

Regular training sessions for interns and staff on data entry standards, reconciliation techniques, and reporting formats (Forms C and B) would strengthen institutional capacity and reduce errors.

7.2.3 Strengthening Professional Values

UCAA's core values customer focus, accountability, teamwork, efficiency, innovation, and integrity should continue to be emphasized in daily operations. Reinforcing these values through mentorship and supervision will ensure that interns and staff uphold professional standards.

7.3 Recommendations to the University

7.3.1 Provision of Templates and Logbooks

The university should provide standardized templates for internship reports and official logbooks. This would guide students in documentation and ensure consistency across submissions.

7.4 Recommendations to Fellow Interns

7.4.1 Open-Mindedness

Interns should approach tasks with flexibility and willingness to learn. Aviation statistics often involve repetitive or complex assignments, and openness ensures growth.

7.4.2 Task Completion and Accountability

Interns should prioritize completing tasks on time and taking responsibility for outputs. This builds trust with supervisors and strengthens professional discipline.

7.4.3 Continuous Learning

Beyond assigned tasks, interns should explore tools such as Python, Power BI, and advanced Excel functions. This proactive learning enhances analytical capacity and prepares them for future roles.

7.4.4 Teamwork and Collaboration

Interns should embrace teamwork, share knowledge with peers, and support each other during shift work. Collaboration ensures accuracy and efficiency in statistical reporting.

Chapter 8 Conclusion

The internship at the Uganda Civil Aviation Authority provided a valuable bridge between academic theory and professional practice. Over the twelve weeks, I gained practical exposure to aviation statistics, strategic planning, and reporting workflows. The experience allowed me to apply technical skills in data collection and verification, while also strengthening professional competencies such as teamwork, communication, and ethical responsibility.

From a technical perspective, I advanced my ability to manage datasets using Excel and Python, particularly through tasks such as analysis of weekly and monthly reports, preparing sector passenger and cargo forms, and generating quarterly traffic reports. These activities reinforced the importance of accuracy and consistency in aviation statistics. Professionally, I developed confidence in collaborative work, learned to adapt to organizational structures, and practiced clear documentation through weekly reports and logbooks.

Academically, the internship validated the relevance of my coursework. Concepts from *Introduction to Data Science* were applied in cleaning flight data, while *Data Storytelling and Visualization* informed the preparation of sector forms and figures that communicated traffic flows effectively. The use of dictionaries in summarizing operator routes demonstrated the practical application of *Data Structures*. These connections highlighted how classroom learning directly supports industry tasks.

Overall, the internship was a transformative experience. It not only enhanced my technical and analytical skills but also clarified my career direction in aviation data analysis and strategic planning. The lessons learned will serve as a foundation for future professional growth, ensuring that I can contribute meaningfully to the aviation sector and beyond.

Chapter A Weekly Reports

This appendix contains selected weekly reports from the internship. They highlight the three core sections of the Strategic Planning Department: Statistics, Planning, and Monitoring & Evaluation. Only representative weeks are included, as many activities were repeated.

Week 1 — Statistics Section

During this week, I was introduced to the Statistics Section. I learned how manifests are collected from handlers and how ADHOCS are standardized before entry. Reconciling discrepancies between handler data and Control Tower records gave me my first exposure to the importance of accuracy in aviation statistics.

- Collecting manifests from handlers (Menzies, DAS, MONUSCO).
- Entering passenger and cargo data into Excel.
- Reconciling discrepancies between handler data and Control Tower records.

This week laid the foundation for understanding aviation data workflows.

Week 6 — Planning Section

This week focused on the Planning Section, particularly updating the business plan. I worked with long-term datasets and budget allocations, linking statistical evidence to strategic objectives. This demonstrated how statistics inform organizational strategy.

- Using long-term datasets (1992–2025) for traffic analysis.
- Analyzing budget allocations to the transport sector.
- Drafting inputs for the FY 2024/25 business plan update.
- Linking statistical evidence to strategic objectives.

Week 8 — Statistics Section (Sector Pax and Forms)

This week emphasized reporting formats and sector analysis. I prepared sector passenger forms to show route-based flows and compiled cargo forms to highlight trade volumes. Exposure to

Form C and Form B broadened my analytical scope from monthly datasets to quarterly and aircraft-specific reporting.

- Preparing sector passenger (pax) forms to show route-based flows.
- Compiling cargo forms to highlight trade volumes.
- Completing Form C (passenger movement by aircraft type).
- Initiating Form B (quarterly passenger flows).

Summary

The selected weekly reports illustrate the diversity of tasks across the Statistics, Planning, and Monitoring & Evaluation sections. Week 1 highlighted foundational statistical workflows, Week 6 demonstrated integration of statistics into strategic planning, Week 8 emphasized advanced reporting formats. Together, they show progression from orientation and routine data entry to strategic contributions, while acknowledging that Monitoring & Evaluation was less emphasized during the internship.

Chapter B Code Listings

This appendix contains selected code snippets used during the internship. They illustrate attempts at automation, data verification, and consolidation using Python and Excel. The listings complement the technical skills discussed in Chapter 4.

Python Scripts

Merging Weekly Excel Reports

```
import glob
import pandas as pd

# Collect all Excel files from the weekly reports folder
files = glob.glob("weekly_reports/*.xlsx")

# Read and concatenate into a single DataFrame
df = pd.concat([pd.read_excel(f) for f in files], ignore_index=True)

# Export consolidated dataset
df.to_excel("monthly_report.xlsx", index=False)
```

Listing B.1: Python script for merging weekly Excel reports

Operator Routes Summary

```
def build_operator_routes(arrivals, departures):
    """
#Build a dictionary mapping each operator to
   their inbound and outbound routes.
#Each route includes origin/destination and passenger/cargo/
   mail totals.
    """
    operator_dict = {}

```

```
    # Inbound (arrivals)
    inbound = (
        arrivals.groupby(["operator","Origin"])[["disembark","cargoin
           ","mailin"]]
        .sum()
        .reset_index()
    )

    # Outbound (departures)
    outbound = (
        departures.groupby(["operator","Destination"])[["Embarking","
           Cargo_out","mail_out"]]
        .sum()
        .reset_index()
    )

    # Loop through all operators automatically
    for op in set(inbound["operator"]).union(set(outbound["operator"
       ])):
        operator_dict[op] = {
            "inbound_routes": inbound[inbound["operator"]==op].
               to_dict(orient="records"),
            "outbound_routes": outbound[outbound["operator"]==op].
               to_dict(orient="records"),
            "summary": {
                "total_inbound_passengers": inbound[inbound["operator
                   "]==op]["disembark"].sum(),
                "total_inbound_cargo": inbound[inbound["operator"]==
                   op]["cargoin"].sum(),
                "total_outbound_passengers": outbound[outbound["
                   operator"]==op]["Embarking"].sum(),
                "total_outbound_cargo": outbound[outbound["operator"
                   ]==op]["Cargo_out"].sum()
            }
```

```
        }
    return operator_dict

routes_dict = build_operator_routes(arrivals, departures)

# Loop through all operators
for operator, data in routes_dict.items():
    print(f"\nOperator:{operator}")
    print("InboundRoutes:", data["inbound_routes"])
    print("OutboundRoutes:", data["outbound_routes"])
    print("Summary:", data["summary"])
```

Listing B.2: Sector Pax Operator Dictionary

Traffic and Punctuality Analysis

```
# Top 10 inbound operators by passenger volume
top_inbound_operator = (
    arrivals.groupby('operator')['disembarking']
    .sum()
    .sort_values(ascending=False)
    .head(10)
)

# Top 10 most frequent operators by flight count
most_frequent_operator = arrivals['operator'].value_counts().head(10)

# Top 10 origins by passenger volume
origin_count = (
    arrivals.groupby('origin')['disembarking']
    .sum()
    .sort_values(ascending=False)
    .head(10)
)

```

```
# Build full datetime for ETA and ATA
arrivals["eta_full"] = pd.to_datetime(
    arrivals["Date"].dt.date.astype(str) + " " + arrivals["eta"].
       astype(str),
    errors="coerce"
)
arrivals["ata_full"] = pd.to_datetime(
    arrivals["Date"].dt.date.astype(str) + " " + arrivals["ata"].
       astype(str),
    errors="coerce"
)

# Calculate delay in minutes
arrivals["arrival_delay_minutes"] = (
    arrivals["ata_full"] - arrivals["eta_full"]
).dt.total_seconds() / 60

# Average delay per operator
operator_delays = (
    arrivals.groupby("operator")["arrival_delay_minutes"]
    .mean()
    .sort_values()
)

print(operator_delays.head(10))   # Top 10 most punctual operators
```

Listing B.3: Analyzing top operators, origins, and punctuality

Excel Formulas

Passenger Difference Check

```
=IF(ABS(B2-C2)>2,"Decline","Approve")
```

Chapter C Figures and Screenshots

This appendix contains full-size figures and screenshots referenced in the main chapters. They provide visual evidence of aviation concepts, datasets, and workflows used during the internship.

C.1 ICAO Regions Map

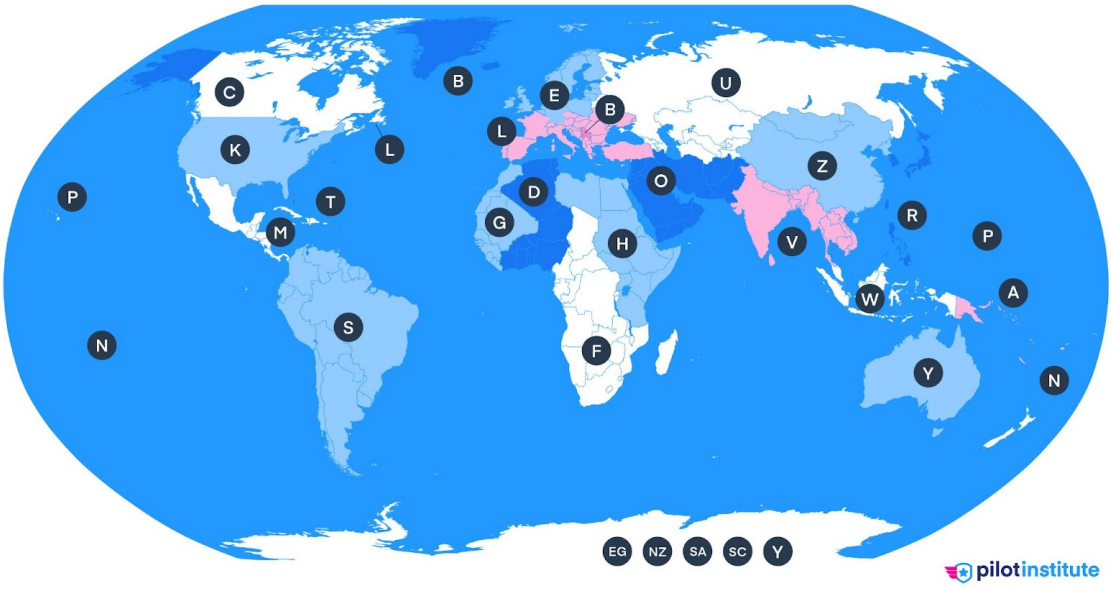
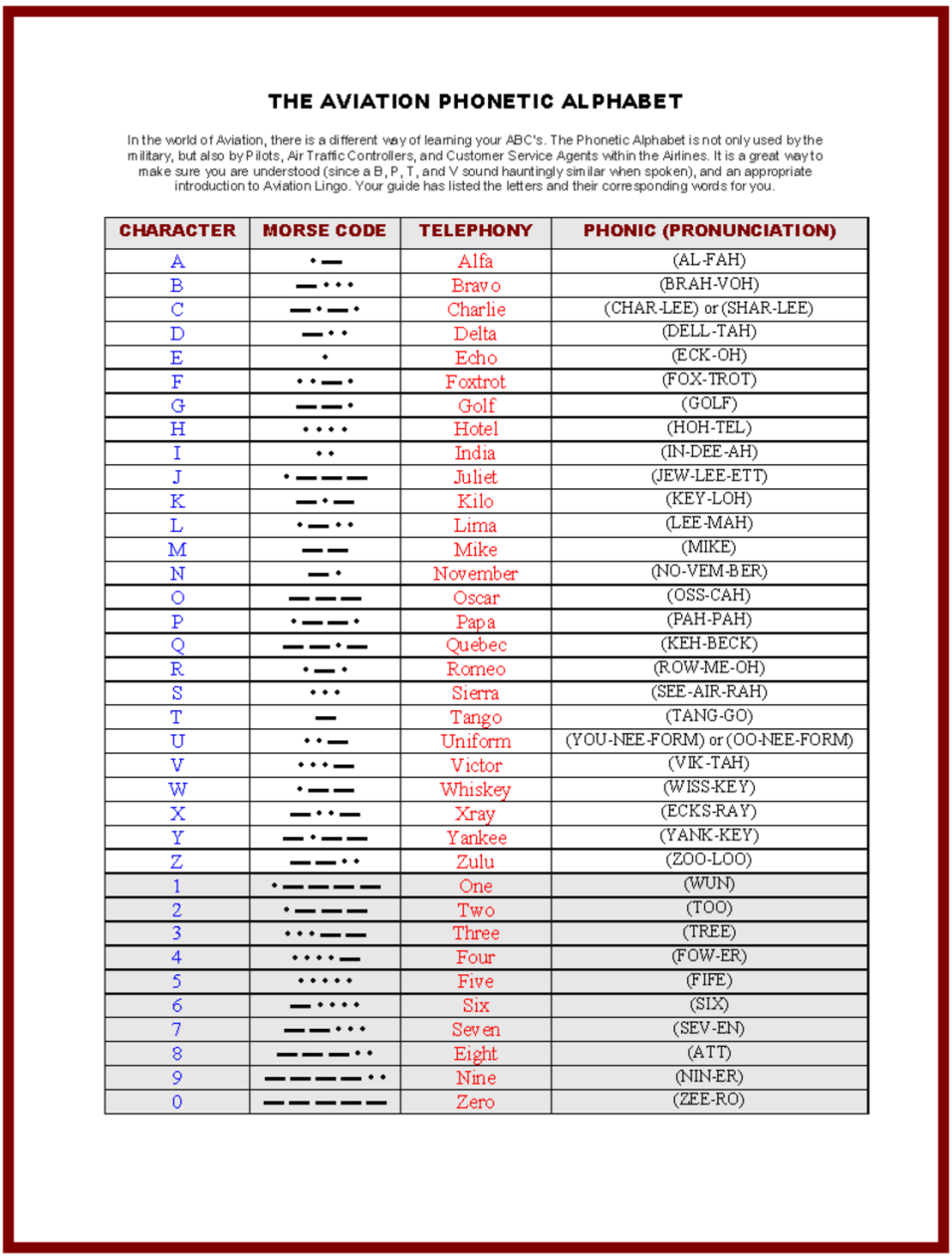


Figure C.1: International Civil Aviation Organization (ICAO) regions.

Source: Pilot Institute (https://pilotinstitute.com/airport-codes-explained/)

C.2 Air Alphabets Chart



THE AVIATION PHONETIC ALPHABET

In the world of Aviation, there is a different way of learning your ABC's. The Phonetic Alphabet is not only used by the military, but also by Pilots, Air Traffic Controllers, and Customer Service Agents within the Airlines. It is a great way to make sure you are understood (since a B, P, T, and V sound hauntingly similar when spoken), and an appropriate introduction to Aviation Lingo. Your guide has listed the letters and their corresponding words for you.

CHARACTER	MORSE CODE	TELEPHONY	PHONIC (PRONUNCIATION)
A	·—	Alfa	(AL-FAH)
B	—···	Bravo	(BRAH-VOH)
C	—·—·	Charlie	(CHAR-LEE) or (SHAR-LEE)
D	—··	Delta	(DELL-TAH)
E	·	Echo	(ECK-OH)
F	··—·	Foxtrot	(FOX-TROT)
G	——·	Golf	(GOLF)
H	····	Hotel	(HOH-TEL)
I	··	India	(IN-DEE-AH)
J	·———	Juliet	(JEW-LEE-ETT)
K	—·—	Kilo	(KEY-LOH)
L	·—··	Lima	(LEE-MAH)
M	——	Mike	(MIKE)
N	—·	November	(NO-VEM-BER)
O	———	Oscar	(OSS-CAH)
P	·——·	Papa	(PAH-PAH)
Q	——·—	Quebec	(KEH-BECK)
R	·—·	Romeo	(ROW-ME-OH)
S	···	Sierra	(SEE-AIR-RAH)
T	—	Tango	(TANG-GO)
U	··—	Uniform	(YOU-NEE-FORM) or (OO-NEE-FORM)
V	···—	Victor	(VIK-TAH)
W	·——	Whiskey	(WISS-KEY)
X	—··—	Xray	(ECKS-RAY)
Y	—·——	Yankee	(YANK-KEY)
Z	——··	Zulu	(ZOO-LOO)
1	·————	One	(WUN)
2	··———	Two	(TOO)
3	···——	Three	(TREE)
4	····—	Four	(FOW-ER)
5	·····	Five	(FIFE)
6	—····	Six	(SIX)
7	——···	Seven	(SEV-EN)
8	———··	Eight	(ATT)
9	————·	Nine	(NIN-ER)
0	—————	Zero	(ZEE-RO)

Figure C.2: Standard aviation phonetic alphabets (Alpha, Bravo, Charlie, etc.).

C.3 Operator Details Output

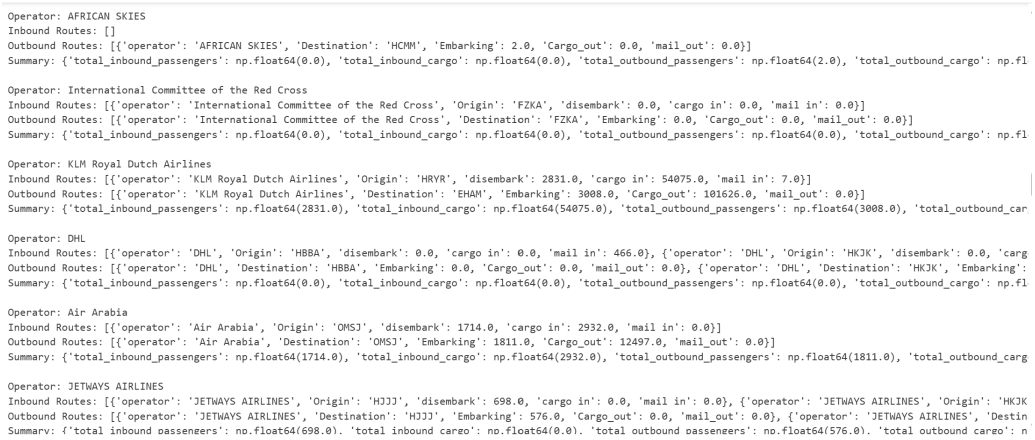


Figure C.3: Operator routes with inbound and outbound cargo and passenger flows.

C.4 May 21-31st analysis

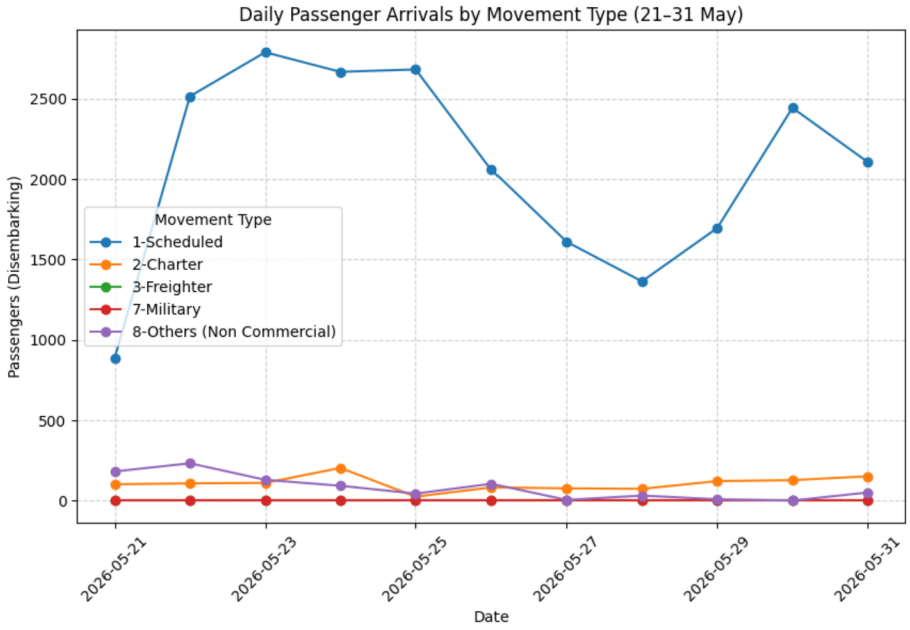


Figure C.4: Disembarking Passengers during May 21-31st.

C.5 May 28-31st analysis

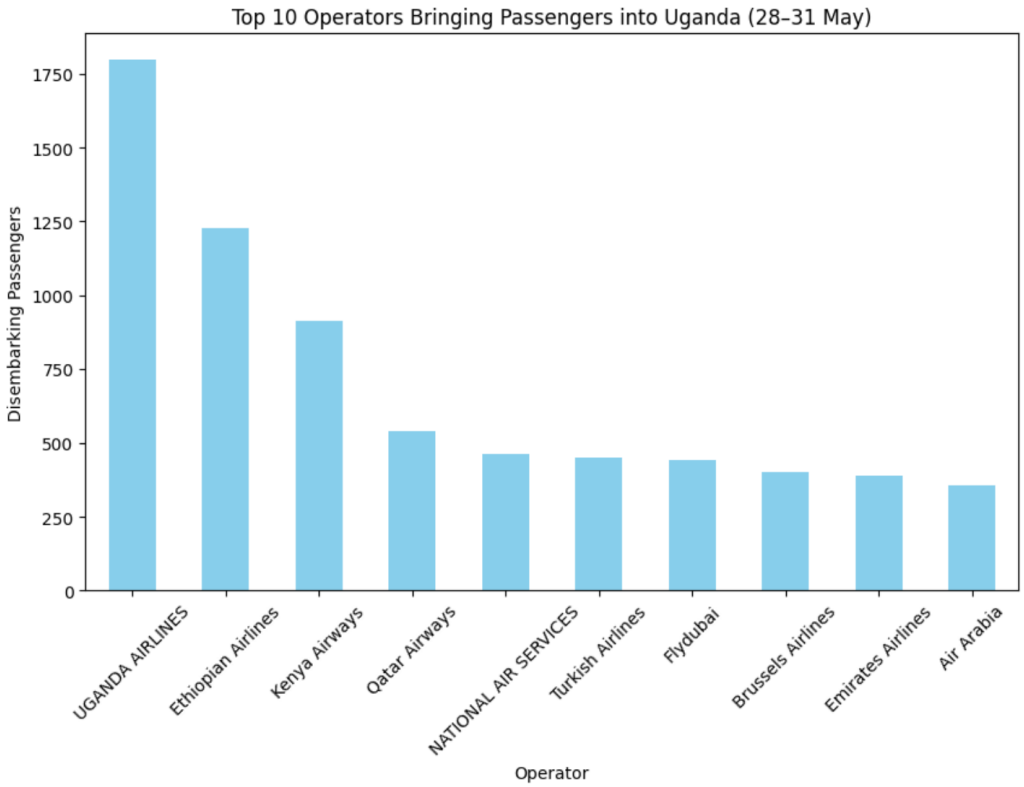


Figure C.5: Disembarking Passengers during May 28-31st.

C.6 Sector Pax

CIVIL AVIATION AUTHORITY
CITY PAIRS 2026

	BEL		KLM		UGD		KQA		AXK		RWD		JMA		UGD	
2024	BRU/EBB	EBB/BRU	AMS/EBB	EBB/AMS	LON/EBB	EBB/LON	NBO/EBB	EBB/NBO	NBO/EBB	EBB/NBO	NBO/EBB	EBB/NBO	NBO/EBB	EBB/NBO	MBO-NBO/EBB	EBB/NBO
JAN	2896	2364	1170	1528	804	2743	5366	6977			223	79			4199	
FEB	2502	3669	1944	2096	718	575	8297	8474				252			3796	
MAR	2283	3881			1622	1471	9908	9757				471			4387	
APR		3530	2831	3008	1046	2209	10894	9527				608		12	5410	
MAY																
JUN																
JUL																
AUG																
SEPT																
OCT																
NOV																
DEC																
TOTAL	7681	13444	5945	6632	4190	6998	33665	34735	0	0	223	1407	0	12	17792	
MKT SHARE	1.11	1.94	0.86	0.96	0.60	1.01	4.86	5.01	-	-	0.03	0.20	-	0.00	2.57	

Figure C.6: Sector Pax template for 2026.

C.7 Form C

FORM C

INTERNATIONAL CIVIL AVIATION ORGANIZATION
AIR TRANSPORT REPORTING FORM
TRAFFIC BY FLIGHT STAGE (TFS)
Scheduled Services (Revenue) - International Operations

Contact _____ **State:** _____
Organization: _____ **Airline:** _____
Tel: _____
Fax: _____ **Year:** _____
E-mail: _____

Stations		Type of aircraft	Number of flights	Capacity available		Revenue traffic		
From	To			Passenger seats (number)	Total payload capacity (tonnes)	Passengers (number)	Freight (tonnes)	Mail (tonnes)
a	**b**	**c**	**d**	**e**	**f**	**g**	**h**	**i**
ABV	EBB	A338		257	45.815			
LOS	EBB	A338		257	45.815			
MED	EBB	A338		257	45.815			
SAA	EBB	A338		257	45.815			
CGO	EBB	A338		257	45.815			
JUB	EBB	A338		257	45.815			

Figure C.7: Form C: Passenger movement by aircraft type

C.8 Form B

Contact person: _____ **State:** _____
Organization: _____ **Year:** _____
Tel: _____ **Air carriers:** _____
Fax: _____
E-mail: _____

Reporting Period (check X)
1st quarter (Jan-Feb-Mar) ☐
2nd quarter (Apr-May-Jun) ☐
3rd quarter (Jul-Aug-Sep) ☐
4th quarter (Oct-Nov-Dec) ☐

Traffic type (check X)
Scheduled ☐
Non-Scheduled ☐

CITY-PAIR		REVENUE TRAFFIC			
From	To	Passengers (Number)	Freight (Tonnes)	Mail (Tonnes)	Air Carriers Included* (by code)
a	*b*	*c*	*d*	*e*	*f*
ABJ	ACC	550	12.738	2.173	
ABJ	ALG	333	2.315	.872	
ABJ	PAR	230	1.234	2.307	
ACC	ABJ	475	10.480	1.735	
ACC	ALG	185	.305	.019	
ACC	PAR	178	.885	.513	
ALG	ABJ	303	1.105	.232	
ALG	ACC	217	0.067	.059	

Remarks (including mention of any unavoidable deviation from instructions):

Figure C.8: Form B: Quarterly passenger flows

Chapter D Business Plan

This appendix contains selected extracts from the updated Business Plan for FY 2025/26. The excerpts illustrate how statistical analysis of air traffic data was integrated into strategic planning and resource allocation at UCAA.

Air Traffic Performance 2024/25

Table D.1: Air Traffic Performance at Entebbe International Airport

S/N	Description	Actual 2018/19	Planned 2024/25	Actual 2024/25	% Actual vs Planned	% Recovery vs 2018/19
1	International Passengers – 7.5%	1,728,799	2,672,030	2,307,246	86.3%	133.5%
2	Domestic Passengers – 3%	27,600	32,956	25,378	77.0%	91.9%
3	Transit Passengers – 7.5%	192,894	298,137	101,261	34.0%	52.5%
4	Imports (tons) – 3%	20,788	24,822	22,597	91.0%	108.7%
5	Exports (tons) – 3%	41,339	49,361	45,329	91.9%	109.6%
6	Commercial Movements – 3%	32,735	39,087	33,251	85.1%	101.6%
7	Overflights – 5%	15,730	21,080	24,695	117.2%	157.1%

Graphical Presentation

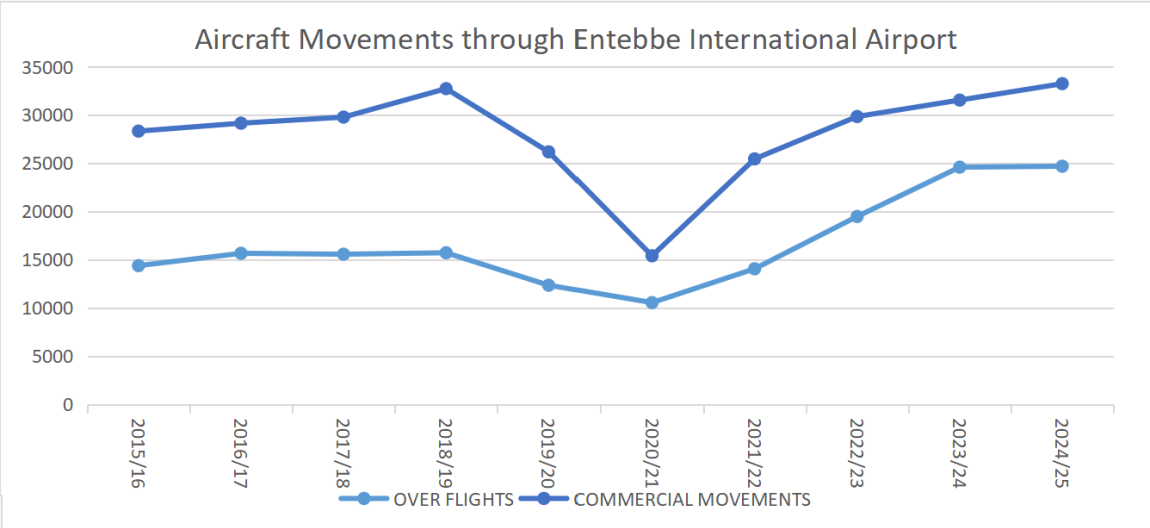


Figure D.1: Aircraft Movements (Overflights vs Commercial Movements, 2015/16–2024/25).

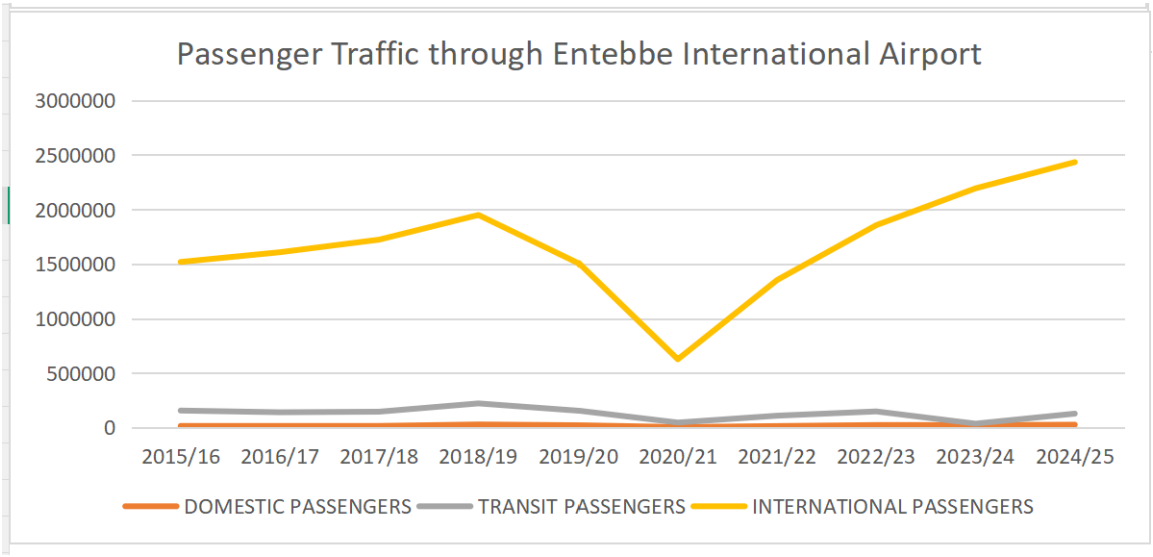


Figure D.2: Passenger Traffic (International, Domestic, Transit, 2015/16–2024/25).

Forecasts

Table D.2: Forecast Summary — projections for FY 2025/26 and 2026/27 based on statistical growth rates applied to 2024/25 actuals

Indicator	Base Year (2024/25 Actual)	Forecast 2025/26	Forecast 2026/27
International Passengers	2,307,246	2,480,312	2,666,335
Domestic Passengers	25,378	26,139	26,923
Transit Passengers	101,261	108,856	117,020
Exports (tons)	45,329	46,689	48,089
Imports (tons)	22,597	23,275	23,973
Commercial Movements	33,251	34,249	35,276
Overflights	24,695	25,930	27,226